56-Raman

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Term Final Examination, Autumn 2025

Course Title: Electronic Devices and Circuits

Course Code: EEE 0714211 / EEE 251

Marks: 50 | **Time: 3*(three)* hours**

(Answer all questions)

Q. No.		Marks
1. a)	**Design** the circuit shown in figure 1. Given $\mu_n C_{ox}$= 100 μm/V^2 ,W= 32 μm, L= 1 μm, I_D= 0.4 mA, V_D= 0.5 V , V_t=0.7 V	[05]

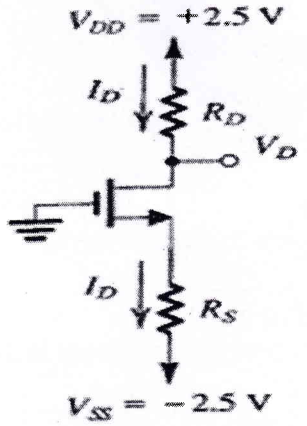


Figure 1: Circuit for 1a

b)	**Find** the range of V_i that will keep the zener diode in "on" state for the circuit shown in figure 2.	[05]

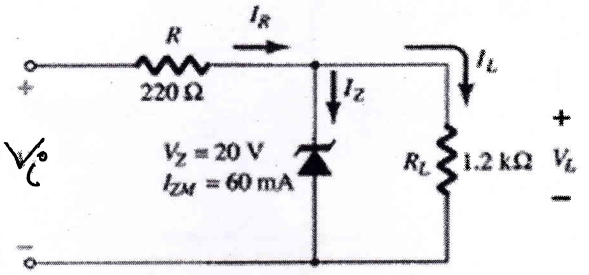


Figure 2: Circuit for 1b

2. a) **Explain** the characteristics and operating principle of n-channel enhancement type MOSFET with necessary diagrams. [05]

b) **Describe** the operating principle of a pnp bipolar junction transistor with necessary diagrams. [05]

3. a) **Design** the circuit shown in figure 3. Let V_t=1V and $k'_n(W/L)$= 1mA/V^2 [04]

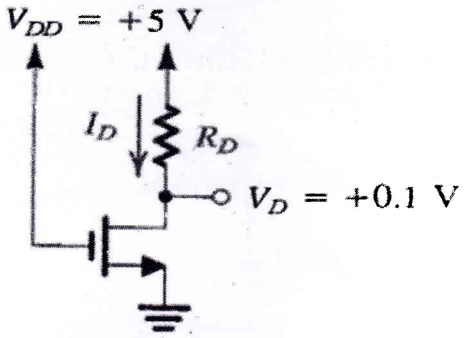


Figure 3: Circuit for 3a

b) For the emitter bias configuration circuit shown in figure 4, **determine** the labeled currents and voltages. [06]

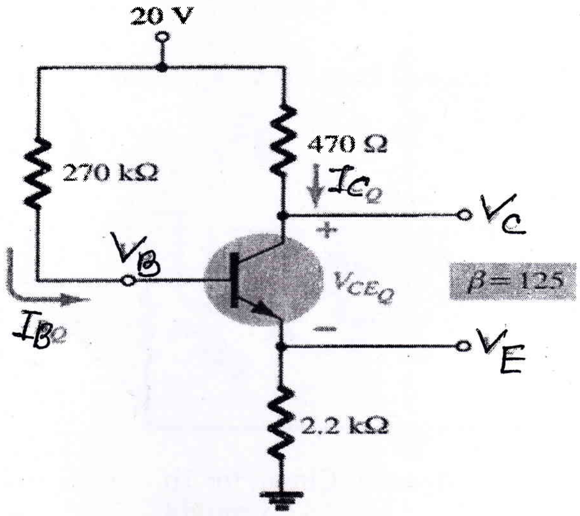


Figure 4: Circuit for 3b

4. a) **Design** a circuit with operational amplifiers that will generate the following equation. Here V_1, V_2, V_3 are the input voltages and V_O is the output voltage. **[06]**

$$v_0 = 5v_1 + 10\frac{dv_2}{dt} + 15\int v_3\,dt$$

b) **Find** the output voltage V_O for the circuit shown in figure 5. **[04]**

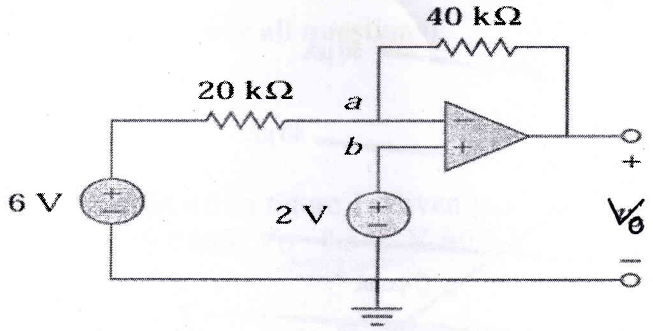


Figure 5: Circuit for 4b

5. a) **Determine** the output voltage V_O for the circuit shown in figure 6. input Voltage V_i is given. **[05]**

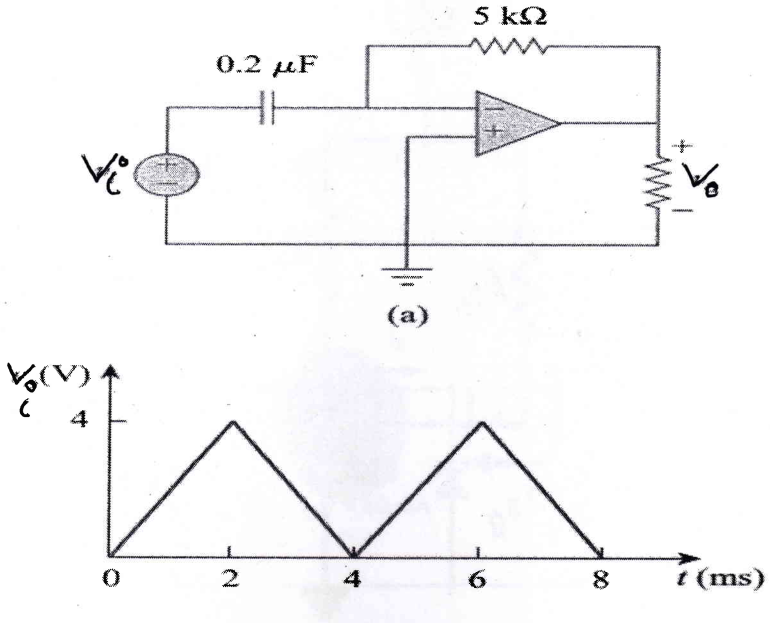
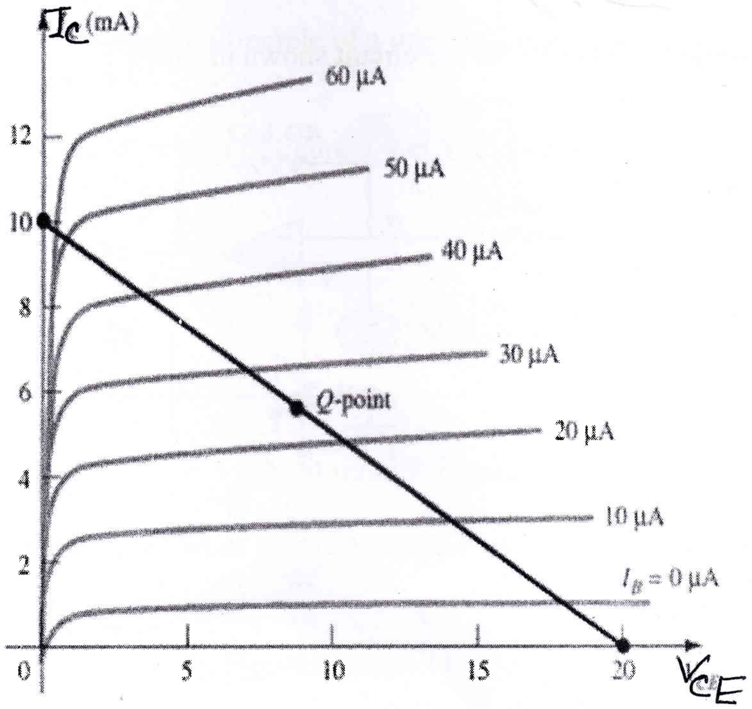


Figure 6: Circuit for 5a

b) **Design** the circuit shown in figure 7 according to the given load line. [05]



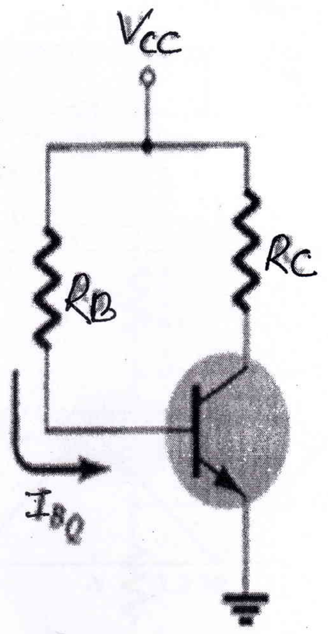


Figure 7: Circuit for 5b